

2.6 Quiz: Motion, Lab Skills, and Spreadsheets

1. Round each value to three significant figures.

(a) 0.006253 m

(c) 9.0048 s

(b) 84.371 kg

(d) 37,820 N

2. Perform each operation and give the answer in scientific notation, rounded to 3 significant figures.

(a) $(5.4 \times 10^3) + (2.8 \times 10^4)$

(b) $(6.20 \times 10^{-2}) \times (5.0 \times 10^3)$

3. Convert each value. Show one line of work using unit factors.

(a) 54.0 km/h to m/s

(b) 25.0 m/s to km/h

4. A student walks along a straight path (forward is +). Start $x_0 = 3.0$ m.

(a) He walks to $x = 19.0$ m. Find displacement Δx .

(b) From $x = 19.0$ m he walks back to $x = 7.0$ m. Find this Δx .

(c) Find total displacement from start to finish.

(d) Find total distance traveled.

Kinematics Equations

Formulas: $v = v_0 + at$ $d = v_0t + \frac{1}{2}at^2$ $v^2 = v_0^2 + 2ad$ $d = \frac{1}{2}(v_0 + v)t$

5. A skateboard accelerates uniformly from rest at 2.0 m/s^2 .

(a) What is its velocity after 3.0 s ?

(b) How far does it travel during this time?

6. A car traveling at 24 m/s applies its brakes and decelerates at 3.0 m/s^2 .

(a) How long does it take for the car to stop?

(b) What distance does the car travel while braking?

7. A ball is thrown vertically upward with an initial velocity of 15 m/s . Use $g = 10 \text{ m/s}^2$.

(a) How long does it take to reach its maximum height?

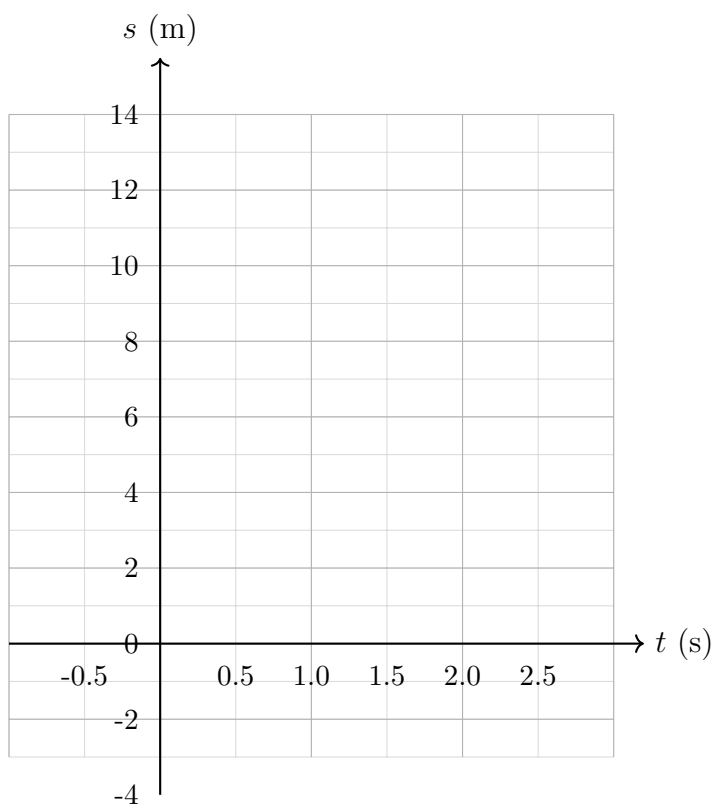
(b) What is the maximum height reached?

Graphing and Data Analysis

A ball is launched upward. The table below shows the ball's position s (in meters) at each time t (in seconds).

Time t (s)	Position s (m)
-0.5	-2.2
0.0	5.0
0.5	9.8
1.0	12.1
1.5	12.0
2.0	9.4
2.5	4.4

8. Plot the data points on the grid below. Draw a smooth curve through the points.



9. A quadratic curve fit to the data gives the equation $s = At^2 + Bt + C$ with the parameters:

$$A = -4.9, \quad B = 12, \quad C = 5$$

Comparing to the kinematic equation $s = \frac{1}{2}at^2 + v_0t + s_0$, answer the following.

- (a) What is the initial position s_0 ?
- (b) What is the initial velocity v_0 ?
- (c) Calculate the acceleration a . (Hint: $A = \frac{1}{2}a$)
10. Find the velocity at $t = 0$ numerically. Use the **average rate of change** (slope) between the points at $t = -0.5$ s and $t = 0.5$ s.

$$v(0) \approx \frac{s(0.5) - s(-0.5)}{0.5 - (-0.5)} =$$

11. Find the velocity at $t = 2.0$ s numerically. Use the average rate of change between $t = 1.5$ s and $t = 2.5$ s.
12. Using your velocity values from problems 10 and 11, calculate the average acceleration over the interval from $t = 0$ to $t = 2.0$ s.

$$a_{\text{avg}} = \frac{v(2) - v(0)}{2.0 - 0} =$$

13. How does this value compare to the acceleration you found in problem 9(c)?